

Matrix multiplication as composition

CHAPTER 4

Matrix multiplication as composition

Two transformations, one after the other.

THE IDEA

Apply one transformation, then another. The combined effect is itself a single transformation, and its matrix is the **product** of the two. That is what matrix multiplication *is* — not an arbitrary rule.

Point	Detail
Order	AB means: apply B first , then A. Read right to left, like $f(g(x))$
Method	Push $\hat{i}$ through both $\rightarrow$ column 1. Push $\hat{j}$ through both $\rightarrow$ column 2
Not commutative	$AB \neq BA$. Rotate-then-shear differs from shear-then-rotate

WORKED EXAMPLE

Matrix multiplication **AB**

apply B first, then A — read right to left

THE MATRICES

$$B = \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix} \quad A = \begin{bmatrix} 0 & 2 \\ 1 & 0 \end{bmatrix} \quad \text{columns tell you where } i \text{ and } j \text{ land}$$

$$B \text{ sends } \begin{bmatrix} i \end{bmatrix} \text{ to } \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{and } \begin{bmatrix} j \end{bmatrix} \text{ to } \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

$$A \text{ sends } \begin{bmatrix} i \end{bmatrix} \text{ to } \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \text{and } \begin{bmatrix} j \end{bmatrix} \text{ to } \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

STEP 1 — follow i (this gives column 1)

$$B \text{ moves } i \text{ to } \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 1 \text{ step along } i, 1 \text{ step along } j$$

now apply A to it :

$$1 \times \begin{bmatrix} 0 \\ 1 \end{bmatrix} + 1 \times \begin{bmatrix} 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

STEP 2 — follow j (this gives column 2)

$$B \text{ moves } j \text{ to } \begin{bmatrix} -1 \\ 0 \end{bmatrix} = -1 \text{ step along } i, 0 \text{ steps along } j$$

now apply A to it :

$$-1 \times \begin{bmatrix} 0 \\ 1 \end{bmatrix} + 0 \times \begin{bmatrix} 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

STEP 3 — the two results become the columns of AB

$$AB = \begin{bmatrix} 2 & 0 \\ 1 & -1 \end{bmatrix} \quad \begin{array}{l} \text{col 1 from } i \\ \text{col 2 from } j \end{array}$$

RULE: push i through both matrices -> column 1. Push j through both -> column 2.

Copy the numbers first. Two matrices means twice as many numbers to mix up — write both matrices' columns on your paper, labelled, before any arithmetic.

TEST — Chapter 4

1. B: $\hat{i} \rightarrow (1,0), \hat{j} \rightarrow (0,2)$. A: $\hat{i} \rightarrow (3,0), \hat{j} \rightarrow (0,1)$. Columns of AB? → **col1 = (3, 0), col2 = (0, 2)**
2. B: $\hat{i} \rightarrow (2,1), \hat{j} \rightarrow (0,1)$. A: $\hat{i} \rightarrow (1,0), \hat{j} \rightarrow (-1,2)$. Columns of AB? → **col1 = (1, 2), col2 = (-1, 2)**
3. B: $\hat{i} \rightarrow (1,-2), \hat{j} \rightarrow (3,0)$. A: $\hat{i} \rightarrow (0,1), \hat{j} \rightarrow (-2,1)$. Columns of AB? → **col1 = (4, -1), col2 = (0, 3)**

FORMULAS FROM THIS CHAPTER

Order

AB = apply B first, then A

Method

push $\hat{i}$ through both $\rightarrow$ col 1; push $\hat{j}$ through both $\rightarrow$ col 2

Warning

$AB \neq BA$